

Reliability Prediction of PV Inverters Based on MIL-HDBK-217F N2

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Abstract — This paper initially discusses the reliability of a 250W Photovoltaic (PV) micro inverter. Using the bill of materials the reliabilities of the main, gate drive, power supply, current and voltage sensing and microprocessor circuits were investigated, and the failure rate and Mean Time Between Failure (MTBF) calculated. The sum of component failure rates equals the complete PV micro inverter failure rate. To account for temperature effects the component failure rate was calculated for each inverter operating temperature, and multiplied by the percentage occurrence of this operating temperature to obtain a weighted failure rate. A similar procedure was used to calculate the failure rate for the main circuits of a 4.6kW & a 4.5kW multi-string inverter. All calculations are based on MIL-17F N2 method.

Index Terms — Failure rate, MIL-HDBK-217F N2, PV micro inverter, PV multi string inverter, Reliability prediction.

I. INTRODUCTION

During the second week of August 2014, the UK solar PV industry achieved 5GW of installed capacity, making the UK the sixth country to reach this capacity. The UK reached this capacity through the adoption of a policy to regulate renewable energy, the Feed in Tariff (FIT) with which PV operators were subsidised [1].

Fig. 1 shows the distribution of the 5GW solar PV generators in the UK [1]. The UK PV generators can be divided into three approximately equal-sized sectors; the first being residential and <5kW. The second sector comprises commercial rooftop or ground systems having a capacity from 5kW-5MW, and the third sector comprises ground mount systems occupying 25-100 acres with installed capacity >5MW.

With this increase in PV renewable energy worldwide there is a respective focus on power electronic converters, because such devices constitute a major part of a PV renewable energy system.

The function of the PV inverter is to convert the solar panel output power from DC to AC efficiently, ensuring the AC has minimum harmonic components and is in phase with the grid supply. A PV inverter is a major component of the PV system, and must have a long operating life, low cost and high efficiency.

To ensure a long PV inverter life, in the design phase it is necessary to assess all potential PV inverter failure modes. A retrospective investigation has been carried out here of the MTBF of common inverter types. The analysis method used involves calculating the failure rate for all inverter

components and summing their individual failure rates to determine the failure rate for the complete inverter.

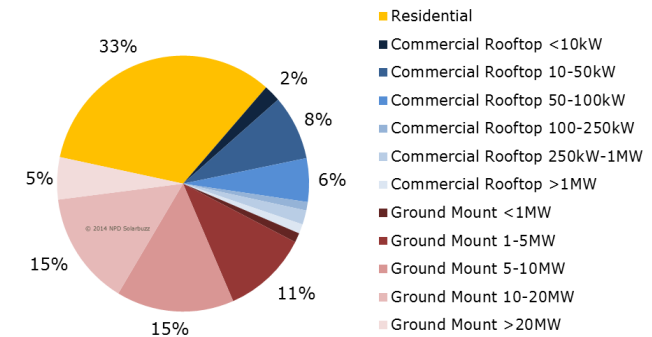


Fig. 1. Application split of 5GW UK solar PV [1].

II. PV INVERTERS

A. Centralized Inverters

Centralized inverters have a high power rating and are connected to an array formed from many PV modules as shown in fig. 2(a).

The array comprises a parallel set of PV panel strings, each producing several hundred volts DC, at around ten Amps. Bypass diodes (D_A in fig. 2(a)) are connected across each panel [2] to prevent excess reverse voltage under mismatch conditions. Each string has a series blocking diode (D_B in fig. 2(a)) to prevent reverse current.

The major disadvantage of this technology is that there can be power losses if a centralized Maximum Power Point Tracking (MPPT) control system is used [3], since a centralized MPPT system optimizes power output from an array and not individual panels. If there are differences in the maximum power point operating voltages and currents for each panel then maximum power can never be obtained with an overall MPPT system. A DC/DC converter providing MPPT can be implemented between each string and the main inverter to ameliorate some of the losses. However, it is still possible for panel losses to occur in a string, due to mismatch losses between PV modules [3]. There are also losses in the series string diodes (D_B).

PV systems using centralized inverters are used only in solar farms or on industrial sites. Being high power, typically $\geq 100\text{kW}$, they supply all 3-phases of the grid.

B. String Inverters

Lower power inverters supplied with a single string of PV panels are known as string inverters. Due to its lower power a string inverter produces a single phase output, as shown in fig. 2(b) [3]. With this PV system, series diodes are unnecessary and MPPT is performed for the whole string. String inverters are frequently found on domestic properties, and are usually roof mounted with rating up to around 100kW [4].

C. Multi String Inverters

In this topology the output of several strings are individually connected to DC/DC converters which provide a higher voltage for a common DC/AC inverter as shown in figure 2(c). Compared with the centralized system, MPPT can be applied to each string individually via the DC/DC converters. Multi string inverter systems are found on commercial properties, and are either roof or ground based.

D. Micro Inverters

With a micro inverter system each PV panel has a single phase inverter rated at around 300W as shown in figure 3(d). The micro inverters are located physically behind their respective PV panel. In this topology every PV panel has its own MPPT control system, and the PV system is easy to enlarge. The micro inverter system is used in domestic PV systems since it offers individual panel MPPT control and is potentially more efficient than a string inverter system. However each inverter is exposed to extreme temperatures, must be waterproof, have a long operating lifetime and be very efficient.

The micro inverter system market is very competitive, many householders nowadays choosing to fit micro inverters instead of string inverters. The major selling features are the promise of a high efficiency, long lifetime and competitive cost. These three factors are difficult for the micro inverter designer to achieve at the same time however.

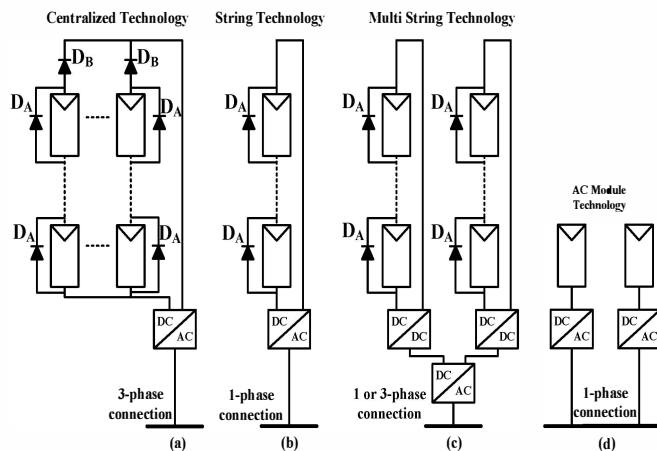


Fig. 2. Overview of PV inverters. (a) Centralized technology. (b) String technology. (c) Multi-string technology. (d) Micro inverter.

As part of an ESPRC sponsored project, WISE PV, EP/K022229/1, an investigation has been carried out into the reliability of PV inverters in general with the aim of predicting usage of PV systems in the future.

III. RELIABILITY PREDICTION METHODS

The term reliability prediction refers to the use of mathematical models and data to estimate the field reliability for systems before empirical data is obtained [5], so as to estimate failure rate over the useful life of the product [6]. Reliability prediction gives an indication of the major decisions that need to be taken in the design stage [7] or later stages such as the development and manufacturing stages [8]. Reliability prediction is based on the assumption that system failure occurs due to a failure of one or several of its component parts. These parts may fail because they are exposed to system stresses [7] or because of incorrect manufacture. Reliability prediction methods are classified into three major groups:

A. Bottom-Up Statistical Methods

These methods are based on statistical curve fitting, the statistical data being collected from the field, from the laboratory or from manufacturers. The methods assume that failure of the system is due to a failure of its components, whose failures are independent from each other. There are two types of BS methods; parts count analysis which assumes that the component operates under typical operating conditions, and parts stress analysis which requires an input of parameters that are included in models of the components' failure rate. This group includes methods such as MIL-217 and RIAC [8].

B. Top-down (TD) similarity analysis methods based on external failure database

These methods use similarity analyses between previous systems and newly designed systems. Not only component failure rates are considered, all failure causes are taken into account, so failure cause analysis is of the utmost importance. This group includes methods such as DERA Transport Reliability Assessment and Calculation System (TRACS) [8].

C. Bottom-up physics-of-failure (BP) methods

These methods need knowledge of the thermal, mechanical, electrical and chemical life cycle environment as well as any processes leading to failures in the field, in order to apply appropriate failure models. For example, the CALCE software package can predict the time to failure for an electrical circuit board or component by targeting the most common failure mechanism at various sites. The methods require information such as material characteristics, geometry, environment and operational loads. This group includes methods such as FIDES [8].

IV. FAILURE RATE OF ELECTRONIC COMPONENTS

Failure rate (λ) is defined as the number of units failing per unit time. Every product has a failure rate, and this failure rate changes throughout the life of the product, producing the 'Bathtub' curve shown in fig. 3 [6].

The Bathtub curve can be divided into three sections. The first is known as the infant mortality period, and the relatively high failure rate in this period is a result of manufacturing errors. The aim of the manufacturer is to eliminate the early failure period so that no product in this period reaches the customer [6]. The second section is known as the normal operation or useful life period, and the manufacturer aims to increase the duration of this period as much as possible (or in some cases extend it to just past the warranty period). The final section is the wear out period and this section occurs beyond the useful life period, when the product becomes old. Failure of the product in this section is due to material degradation, oxidation, vibration, or temperature and chemical damage. The infant mortality period and useful life period are covered by the manufacturer's warranty.

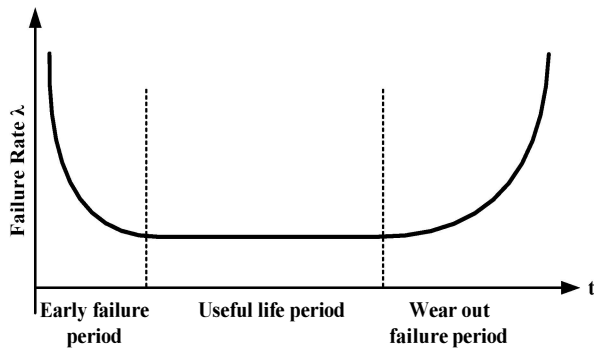


Fig. 3. Bathtub Curve.

The reciprocal of failure rate is known as the Mean Time Between Failure (MTBF), and this term is often applied to products that can be repaired and returned to service. While Mean Time To Failure (MTTF) was frequently used to measure the time to failure for non-repairable products, in practice MTBF is used for both repairable and non-repairable [9].

From the above, it can be seen that several methods are used to predict reliability. Amongst all these methods MIL-217F N2, developed by the US Department of Defense is internationally used to calculate the failure rate of electronic components used in military and civil equipment. This method has also been used here because it is freely available and is frequently used commercially. As shown in table III, failure rate equations in MIL-217F N2 include several factors such as temperature, environment, quality and voltage stress ratio [10]. MIL-217F N2 offers a part count method for most electronic components and a part stress method for other

components such as microcircuits (operational amplifiers and microprocessors).

TABLE I
FAILURE RATE EQUATIONS FOR CAPACITORS AND MOSFETS [10]

Component	Failure rate (Failures 10^6 hours)	
Capacitors	$\lambda_p = \lambda_0 \pi_T \pi_C \pi_V \pi_{SR} \pi_Q \pi_E$	λ_0 = constant depending on capacitor type $\pi_T = \exp\left(\frac{-E_a}{8.617 \cdot 10^{-5} \left(\frac{1}{T+273} - \frac{1}{298}\right)}\right)$ $\pi_V = \left(\frac{S}{0.6}\right)^y + 1$, $S = \frac{\text{Operating Voltage}}{\text{Rated Voltage}}$ $\pi_C = C^x$
MOSFET	$\lambda_p = \lambda_0 \pi_T \pi_A \pi_Q \pi_E$	$\lambda_0 = 0.012$ $\pi_T = \exp\left(-1925 \left(\frac{1}{T_j+273} - \frac{1}{298}\right)\right)$
E_a : activation energy, x and y depend on capacitor type, T : ambient temperature, T_j : junction temperature, p : rated resistor power		

The failure of the whole system is given by the sum of the failure rates for all the system components, as shown in equation 1.

$$\lambda = \sum_{i=1}^n N_i \lambda_i \quad (1)$$

Where:-

λ : Failure rate for the whole system

n : number of component categories

N : quantity of i^{th} component

λ_i : Failure rate of i^{th} component

When designing PV inverters, Aluminum Electrolytic Capacitors are sometimes chosen to store DC energy. An alternative capacitor type that the designer may choose is the metal film capacitor.

As an example, to highlight the difference between these two types, fig. 4 depicts the failure rate of a 22 μ F, 450V, Aluminum Electrolytic capacitor versus voltage stress and ambient temperature, and fig. 5 depicts the failure rate for a Metallized Film capacitor. The failure rate equation for a capacitor is shown in table I; λ_0 is the base failure rate, π_V is the voltage stress ratio factor, π_T is the temperature factor, π_C is the capacitance factor (failure rate is calculated based on the value of the capacitor in μ F), π_{SR} is the series resistance factor and is defined as the ratio between effective resistance voltage drop between capacitor and power supply to the voltage applied to the capacitor, π_E is the environment factor (the environment is considered 'ground fixed' for all components), and π_Q is the quality factor. Figures 4 and 5 show failure rates for quality factors equal to 1 and 10. It is clear that failure rate increases with ambient temperature and voltage stress ratio when the quality factor is equal to 1 or 10. But the failure rate of the Metallized Film capacitor is lower than the failure rate

of the Aluminum Electrolytic capacitor by a factor of around 2. To reduce the failure rate, the capacitors can be voltage derated. It is usual to derate the capacitor voltage to at most 75% of rated. In some cases this is achieved by connecting equal value capacitors in series to share the voltage. When capacitors are connected in series then care must be exercised to ensure they share voltage over all conditions and over the inverter lifetime.

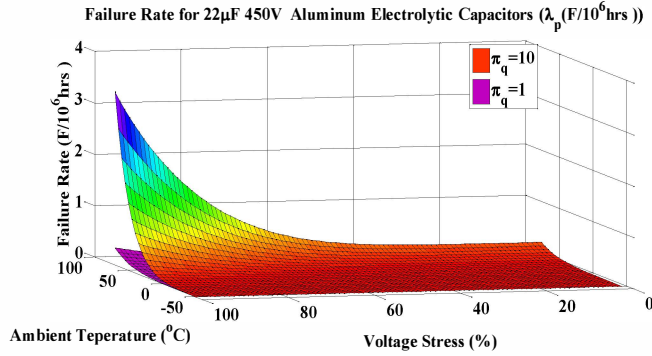


Fig. 4. Al. Elec. capacitor failure rate as a function of Quality factor, ambient temperature and voltage stress.

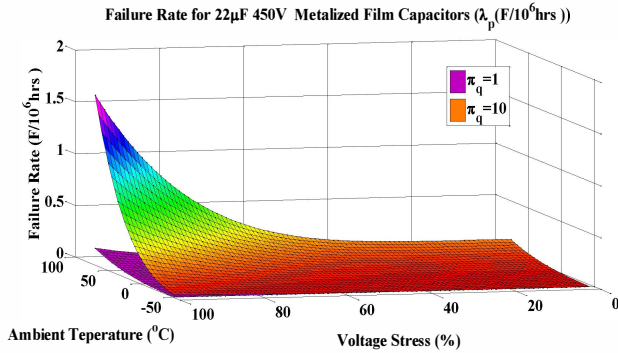


Fig. 5. Metallized film capacitor failure rate as a function of Quality factor, ambient temperature and voltage stress.

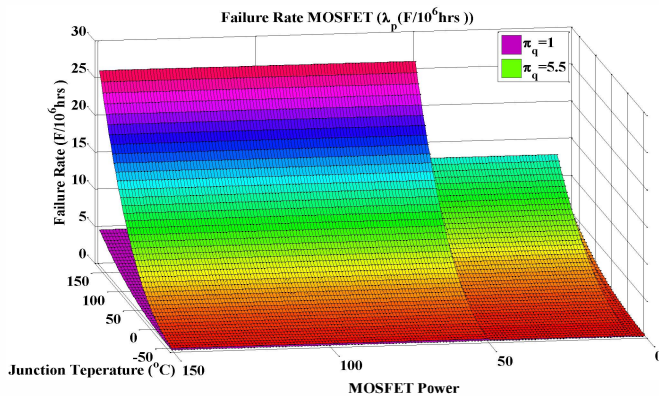


Fig. 6. Transistor failure rate as a function of quality factor, power rating and junction temperature.

N-channel MOSFET switching transistors are used in lower power inverters. The failure rate equation for the transistors is

shown in table I; λ_b is the base failure rate. π_T is temperature factor as a function of junction temperature, π_A is the application factor which depends on the power of the transistor, π_E is the environment factor and the environment is considered 'ground fixed', and π_Q is the quality factor. Fig. 6 depicts the failure rate of a transistor as a function of power and junction temperature, when the quality factor equals 5.5 and 1.

IV. PV MICRO INVERTER FAILURE RATE

To calculate the failure rate of a PV micro inverter, a 250W PV micro inverter [11] was taken as an example. The main circuit for this micro inverter is shown in fig. 7. In determining the micro inverter failure rate all components in the bill of materials were considered, not just the main circuit shown in fig. 7, but also the gate drive circuits, power supply circuits, sensing circuits, microprocessor circuit, etc.

For each component in the micro inverter, the failure rate was calculated from the equations shown in table I. Initially the ambient temperature was set at 20°C and the sum of the component failure rates is then equal to the PV micro inverter failure rate, as given in equation 1. The MTBF was calculated from the inverse of failure rate as shown in equation 2.

$$MTBF = \frac{10^6 \text{ hours}}{\lambda_p} \quad (2)$$

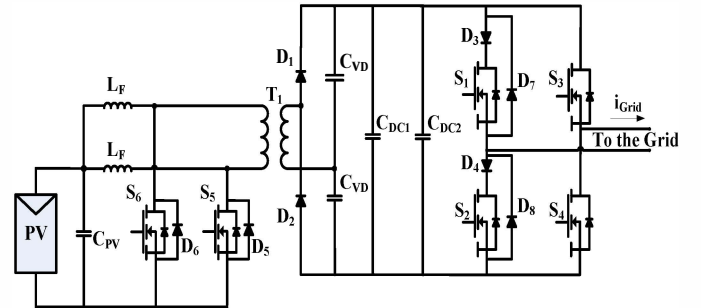


Fig. 7. 250W PV micro inverter main circuit in [11].

So as to more accurately calculate the micro inverter failure rate and MTBF, the failure rate for each component can be calculated for a particular temperature, and then multiplied by the percentage occurrence of this temperature.

As an example of this process, the hourly temperature has been recorded for Manchester, UK, by the Whitworth Meteorological Observatory for the years 2010-2014. Fig. 8 shows a histogram of temperature versus percentage time spent at that temperature over this four year period. Using this histogram the failure rate can be calculated for each component at each temperature. The failure rate at each temperature for the micro inverter is then found from

equation1 and multiplied by the percentage time spent at that temperature to produce a weighted failure rate. Lastly, MTBF is calculated from the weighted failure rate.

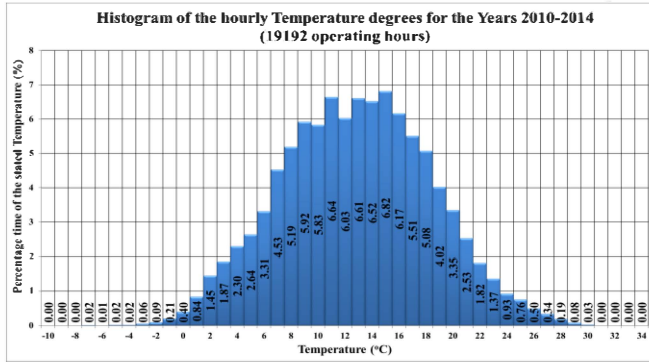


Fig. 8. Temperature histogram for Manchester city for the years 2010-2014.

Table II shows the failure rate for the PV micro inverter components at 20°C when π_q =low. The same calculation was made for all other temperatures when π_q =1 and π_q =low to provide an un-weighted and weighted Failure Rate for the PV micro inverter.

TABLE II
UN-WEIGHTED FAILURE RATE (F/10⁶HRS) AT 20°C WHEN
QUALITY FACTOR LOW

Un Weighted Failure Rate at 20°C (F/10 ⁶ hrs) when π_q =low	
Capacitors	λ_p =23.1103
Diodes	λ_p =02.0816
Inductors & Transformers	λ_p =00.1843
Transistors	λ_p =22.3440
Resistors	λ_p =09.4170
Microcircuits	λ_p =00.4752
Optocouplers & Relays	λ_p =00.1665
Connectors	λ_p =02.8427
λ_{pt}(F/10⁶hrs)	60.62

Table III summarises the total failure rate and MTBF of the PV micro inverter for the temperature distribution of fig. 9; table III(a) shows the results when π_q =1, and table III(b) shows the results when π_q =low.

TABLE III
TOTAL FAILURE RATE AND MTBF OF PV MICRO INVERTER

Total failure rate of PV micro inverter when π_q =1	
Weighted Failure Rate of PV Micro Inverter (λ_p (F/10 ⁶ hrs))	8.069
Mean Time Between Failures (MTBF) (hours)	123938.39
(a)	
Total failure rate of PV micro inverter when π_q =low	
Weighted Failure Rate of PV Micro Inverter (λ_p (F/10 ⁶ hrs))	50.76
Mean Time Between Failures (MTBF) (hours)	19699.69
(b)	

From the above calculations, the PV micro inverter main circuit failure rate was found to be 19.85 F/10⁶ hours and the

MTBF to be 50369 hours when the quality factor is low.

The analysis shows that in the PV micro inverter main circuit, MOSFETs have the highest failure rate and the failure rate depends on the power harnessed by the PV panel and the ambient temperature. Aluminium Electrolytic capacitors and Metallized Polypropylene Film capacitors have acceptable failure rates if they are derated in voltage by nearly 50% (e.g. the failure rate of a 22μF, 450V Aluminum Electrolytic Capacitor is 0.133 F/10⁶ hours at 50°C and quality factor low when it is derated by 50%, and the failure rate is 1.312 F/10⁶hours for the same capacitor at the same conditions when it is derated by 50%). The failure rate for the capacitors also depends highly on ambient temperature.

The other components in the bill of materials have low failure rates, but the complete micro inverter is estimated to have a failure rate between 2.25 and 14.15 years as the quality factor changes.

IV. PV MULTI STRING INVERTER FAILURE RATE

The power circuits of two typical PV multi-string inverters are taken next as an example. The first circuit is shown in fig. 9 [12]. The inverter can be connected to two strings, each of which has its own MPPT system comprising a boost converter. Each string is decoupled by a 15μF film capacitor, C_{PV} [13]. The capacitors C_{Grid} in the half bridge inverter are each comprised of five, 1000μF, 450V, Aluminum Electrolytic capacitors [13] connected in parallel. The voltage across each capacitor C_{Grid} is at least equal to the peak grid voltage.

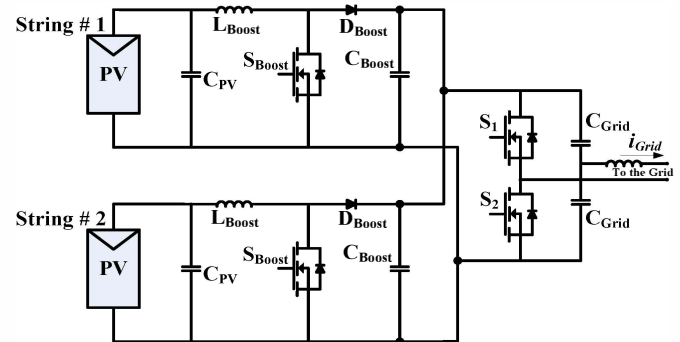


Fig. 9. First multi string inverter main circuit [12].

The second multi-string inverter considered is shown in fig. 10 [3]. This inverter can be connected to three strings, the input voltage range for each string being 200V-500V. The DC-DC converter in each string has a high frequency transformer allowing the PV panels to be grounded [3]. No details are given for the capacitors C_{PV}, so it is assumed here that they are the same as those in the first multi string inverter. The decoupling capacitors, C_{DC}, are each comprised of three, 310μF, 400V, Aluminum Electrolytic capacitor [3] in series. The three string boost converter outputs are parallel connected

to a single-phase, two-level inverter to inject power into the grid.

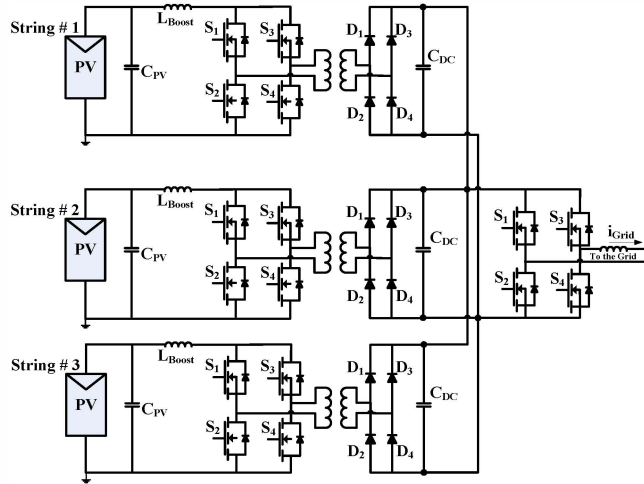


Fig. 10. Second multi string inverter main circuit [3].

A calculation was undertaken of the failure rates for the two inverter systems. The failure rates for each were calculated at the temperatures in the histogram of fig. 8, when π_q =low, to provide un-weighted and weighted Failure Rate results.

Table IV(a) shows the total failure rate and MTBF of the first inverter when π_q =low; and table IV (b) shows the total failure rate and MTBF of the second inverter when π_q =low.

The failure rates given in table IV are calculated for the main power circuits of the two multi string inverters alone, since details of the subsidiary circuits were not available. MOSFETs have the highest failure rate and the failure rate depends on the power harnessed by the strings and the ambient temperature.

TABLE IV

TOTAL FAILURE RATE AND MTBF OF FIRST AND SECOND INVERTERS

Total failure rate of the first multi string inverter when π_q =low	
Weighted Failure Rate of PV Micro Inverter ($\lambda_p(F/10^6\text{hrs})$)	18.66
Mean Time Between Failures (MTBF) (hours)	53594
(a)	
Total failure rate of the second multi string inverter when π_q =low	
Weighted Failure Rate of PV Micro Inverter ($\lambda_p(F/10^6\text{hrs})$)	20.71
Mean Time Between Failures (MTBF) (hours)	48279
(b)	

IV.CONCLUSION

Failure rate and MTBF are used as reliability metrics for electronic components and for systems. The variation of component failure rate has been analyzed for a single micro inverter and two typical multi string inverters operating over a wide variation in ambient temperature and voltage stress ratio. The failure rate for the micro inverter has been calculated for different quality factors based on the MIL-217F N2 method,

and the failure rates for multi-string inverters have been calculated for low quality factors.

The MTBF for the micro inverter analyzed is estimated at 123938 hours (14.15 years) for a high quality system, whereas it is 19699 hours (2.25 years) for a low quality system. Unfortunately, there is no any evidence to prove that PV micro inverters are failing at these rates, because manufacturers do not provide data on the failure rate of their micro inverters.

The failure rates are calculated for the main power circuits of the two multi string inverters alone, since details of the subsidiary circuits were not available. The MTBF for the first multi string inverter analyzed is estimated at 53594 hours (6.1 years), whereas it is 48279 hours (5.5 years) for the second multi string inverter at low quality factor.

The UK mean temperature from fig. 9 is around 14°C, and MTBFs are estimated in a weighted manner from this figure. The temperature inside the PV inverter box will be higher than the ambient temperature. Also, in many places where the insolation and temperature are higher than in the UK, the reliability of the PV inverters will be lower. Several items are not considered in the calculations such as soldered joints, seals, main board, wiring or thermal dissipation system (e.g. thermal compound which can dry out or thermal pads which can degrade through thermal cycling).

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